

Lecture 7

Linked list Deletion and more

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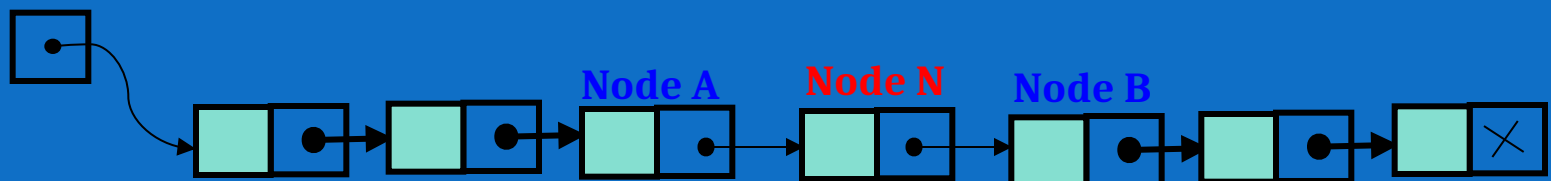
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Deletion from a Linked List

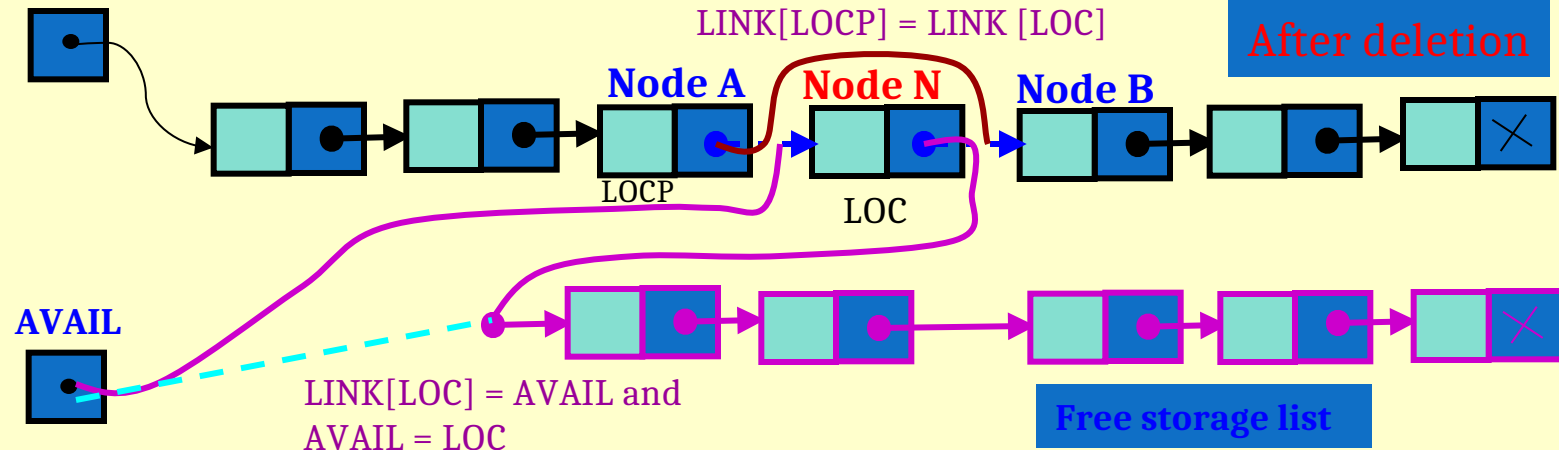
The next pointer field of node-A now points to node-B, where node-N previously pointed
The next pointer field of N now points to the original first node in the free pool, where AVAIL previously pointed.
AVAIL now points to the deleted node-N

START



(a) Before deletion of **Node N**

START



Finding a Node and Delete

Step 1. Underflow ,IF(START = NULL)

write : Underflow and EXIT.

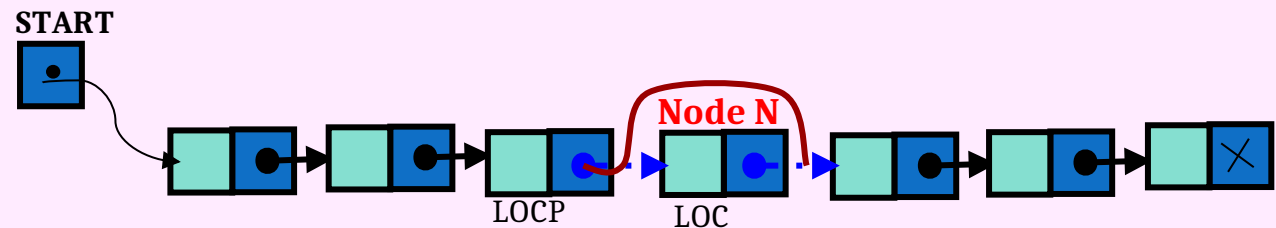
Step 2. Repeat step 3 until NODE[DATA] = ITEM

Step 3. if NODE[DATA] = ITEM

PREVIOUS_NODE[LINK] = NODE[LINK]

Step 4. EXIT

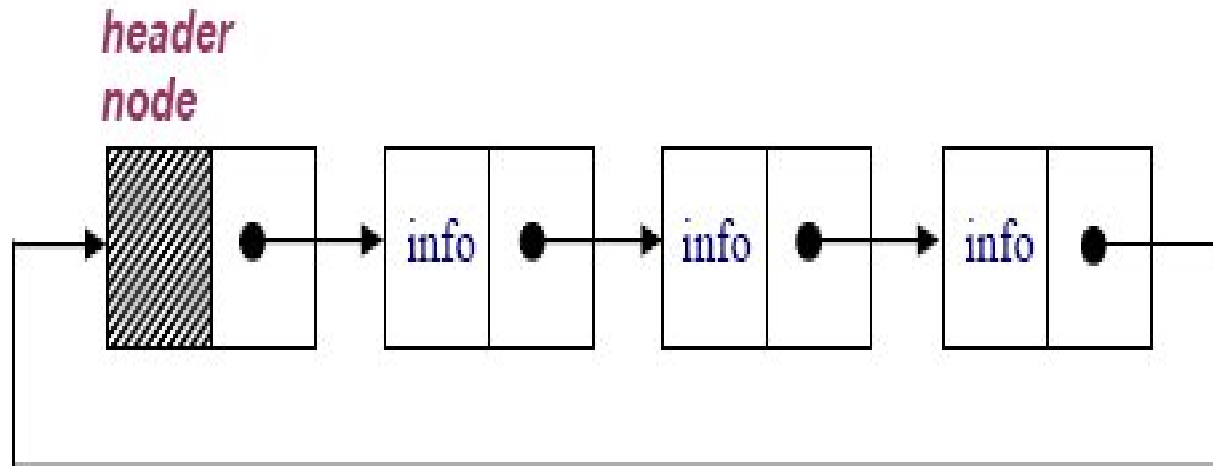
(Think how you will preserve **PREVIOUS_NODE[LINK]**)



Variations of Linked Lists

- *Circular linked lists*

- The last node points to the first node of the list
- How do we know when we have finished traversing the list? (Tip: check if the pointer of the current node is equal to the head.)



a circular linked list with a header node

Variations of Linked Lists

circular linked list:

In a circular linked list there are two methods to know whether a node is the first node or not.

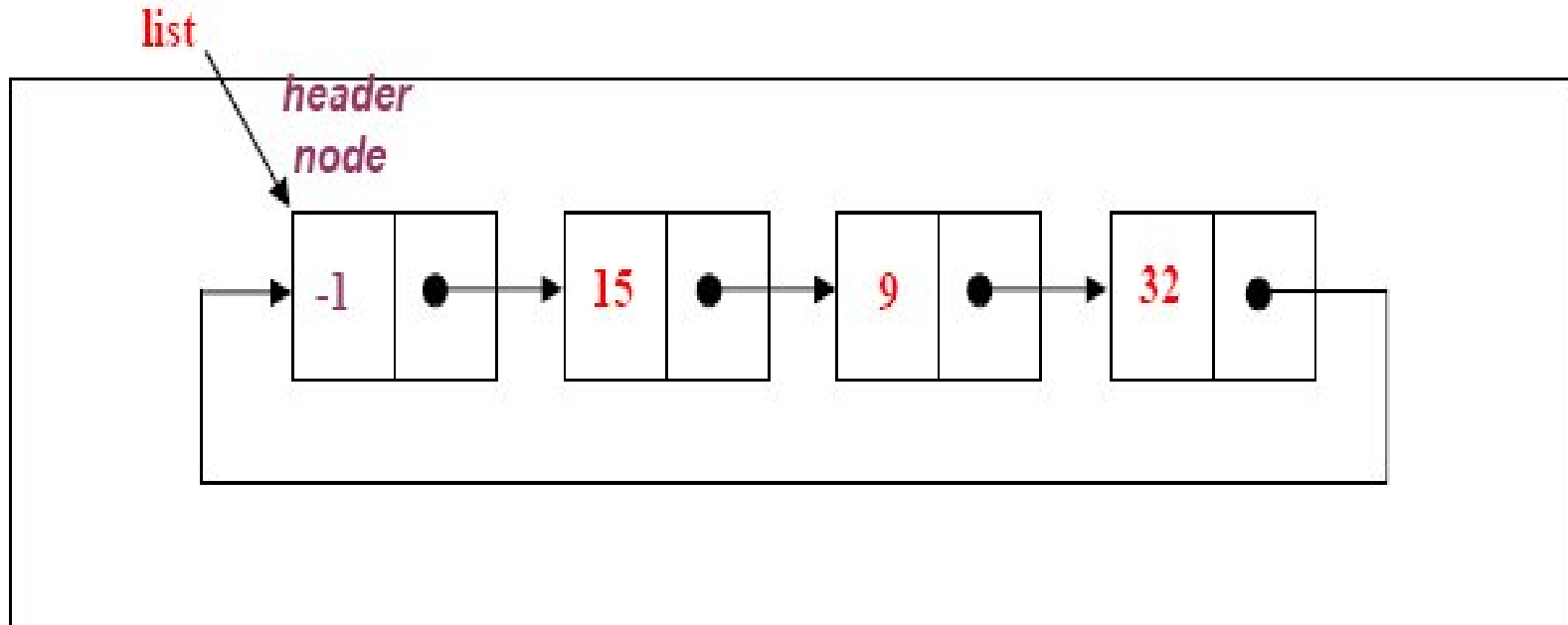
Either an external pointer, ***list***, points the first node or

A ***header node*** is placed as the first node of the circular list.

The header node can be separated from the others by either having a ***sentinel value*** as the info part or

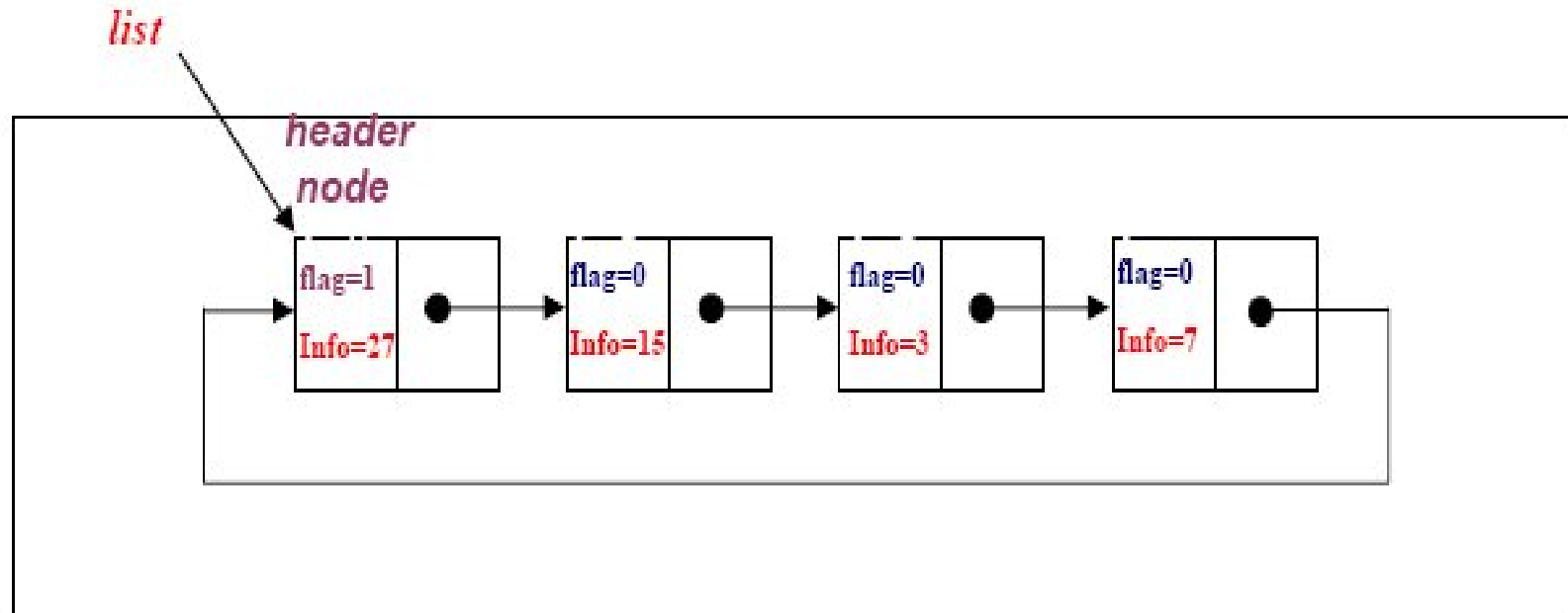
having a dedicated ***flag*** variable to specify if the node is a header node or not.

CIRCULAR LIST with header node



- The header node in a circular list can be specified by a ***sentinel value*** or a dedicated ***flag***:
- ***Header Node with Sentinel***: Assume that info part contains positive integers. Therefore the info part of a header node can be -1.

CIRCULAR LIST with header node



- **Header Node with Flag:** In this case an extra variable called flag can be used to represent the header node.
- For example flag in the header node can be 1, where the flag is 0 for the other nodes.

Example of application of circular linked list

A good example of an application where circular linked list should be used is a timesharing problem solved by the operating system.

In a timesharing environment, the operating system must maintain a list of present users and must alternately allow each user to use a small slice of CPU time, one user at a time.

The operating system will pick a user, let him/her use a small amount of CPU time and then move on to the next user, etc.

For this application, there should be no NIL pointers unless there is absolutely no one requesting CPU time.

Advantages

Each node is accessible from any node.

Disadvantage:

Danger of an infinite loop !

Linear linked list vs Circular Linked Lists

In **linear linked** lists if a list is traversed (all the elements visited) an external pointer to the list must be preserved in order to be able to reference the list again.

Circular linked lists can be used to help the traverse the same list again and again if needed. A circular list is very similar to the linear list where in the circular list the pointer of the last node points not NULL but the first node.

Array versus Linked Lists

Linked lists are more complex to code and manage than arrays, but they have some distinct advantages.

Dynamic: a linked list can easily grow and shrink in size.

We don't need to know how many nodes will be in the list. They are created in memory as needed.

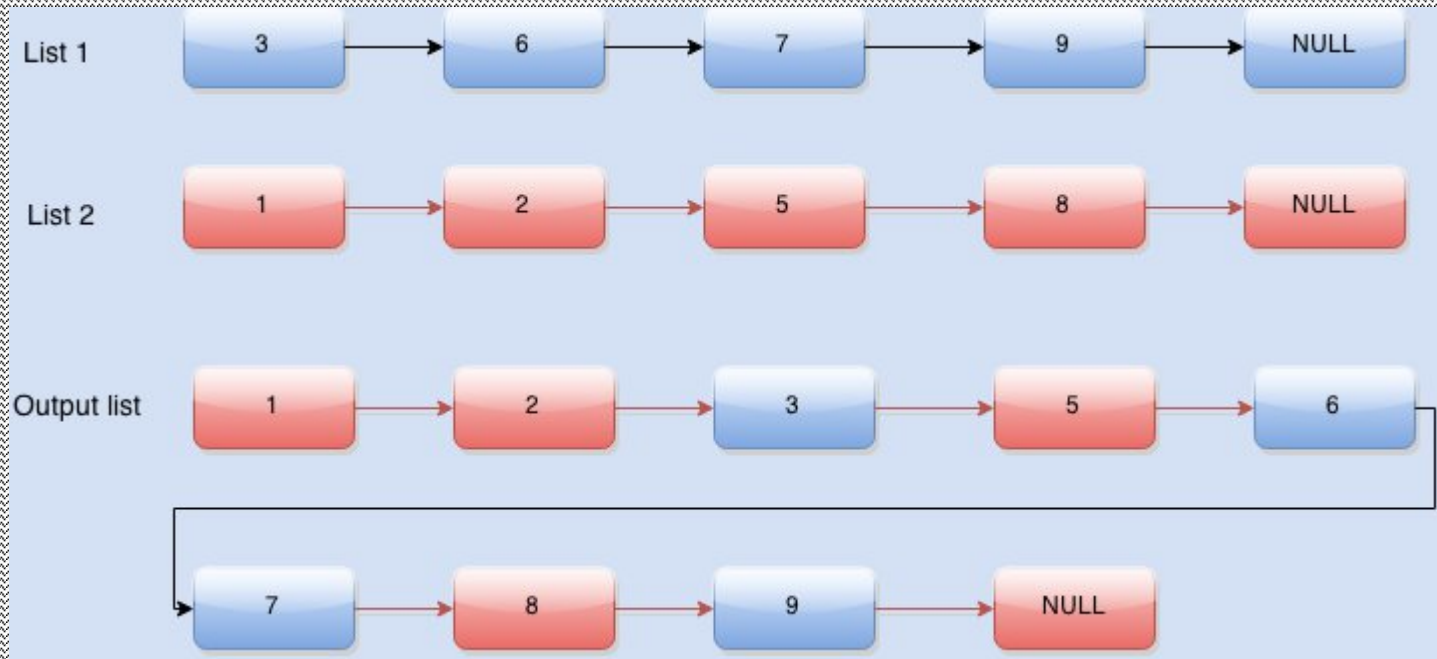
In contrast, the size of a C++ array is fixed at compilation time.

Easy and fast insertions and deletions

To insert or delete an element in an array, we need to copy to temporary variables to make room for new elements or close the gap caused by deleted elements.

With a linked list, no need to move other nodes. Only need to reset some pointers.

Link List Merge



Search

Question?